Here's a **6-month roadmap to become an AI engineer**, with only **free resources**. We'll go step by step, stacking skills in the right order so you don't get overwhelmed.

Month 1: Core Foundations

Goal: Build strong math + Python skills, since they're the backbone of AI.

- **Math:** Linear Algebra, Probability, Calculus basics.
  - Resource: Khan Academy – Linear Algebra
  - Resource: Probability & Statistics
- **Python for AI:**
  - Resource: Python for Everybody (Coursera, free audit)
  - Practice: LeetCode Easy problems (just basics, focus on confidence).

Month 2: Data & Machine Learning Basics

Goal: Learn how data flows through ML models.

- **Data Handling:**
  - Resource: Pandas Data Science Handbook (free)
- **Machine Learning Essentials:**
  - Resource: Andrew Ng's ML course (Coursera, free audit)
  - Resource: Hands-on ML with Scikit-learn & TensorFlow notebooks (GitHub)

Project: Predict house prices with Scikit-learn.

Month 3: Deep Learning Foundations

Goal: Understand neural networks, backpropagation, and training basics.

- **Neural Nets:**
  - Resource: DeepLearning.AI TensorFlow Developer (Coursera, free audit)
  - Resource: Fast.ai Deep Learning for Coders
- **Key Concepts:** Activation functions, loss functions, gradient descent, overfitting, regularization.

Project: Build an image classifier (MNIST or CIFAR-10).

Month 4: Applied AI – NLP and CV

Goal: Dive into practical AI subfields.

- **Natural Language Processing:**
  - Resource: Hugging Face Course
- **Computer Vision:**
  - Resource: PyTorch Tutorials – Vision
- **Extra:** Stanford CS231n (YouTube lectures)

Project: Sentiment analysis on tweets + Image classification with transfer learning.

Month 5: Advanced AI Topics + Deployment

Goal: Go from model training → real-world deployment.

- **MLOps basics:**
  - Resource: Made With ML – MLOps
- **Model Deployment:** Flask/FastAPI + Docker basics.
  - Resource: Full Stack Deep Learning Deployment Module
- **Experiment tracking:** Intro to MLflow, Weights & Biases (both free tiers).

Project: Deploy a small NLP or CV model on Hugging Face Spaces or Streamlit.

Month 6: Specialization + Portfolio

Goal: Showcase your skills with projects that stand out.

- **Choose a track:**
  - Generative AI (LLMs, diffusion models) → Hugging Face Transformers
  - Reinforcement Learning → Spinning Up in Deep RL by OpenAI
- **Portfolio Building:**
  - 3–4 solid projects on GitHub with clear READMEs.
  - Write short LinkedIn/blog posts explaining your work (shows communication skills).

Capstone Project: End-to-end pipeline → Data preprocessing → Model → Deployment.
Example: AI chatbot, medical image classifier, or recommender system.

Tips to Stay on Track

- **Daily practice:** At least 1–2 hours coding, 30 mins theory.
- **Community:** Join Discord/Reddit ML groups, Kaggle competitions.
- **Consistency > Perfection:** Don't get stuck on one concept too long.